

Arman Zarei

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EDUCATION

- **University of Maryland, College Park** Maryland, USA
Ph.D. in Computer Science - Advised by Prof. Soheil Feizi
GPA: 4.0/4.0
Aug. 2023 - Present
- **Sharif University of Technology** Tehran, Iran
B.Sc. in Computer Engineering
Sep. 2018 - Feb. 2023

RESEARCH INTERESTS

Generative Models, Image Generation & Editing, Autoregressive Video Diffusion Models, Computer Vision

RESEARCH EXPERIENCE

- **University of Maryland, College Park** Maryland, United States
Research Assistant - Supervisor: Prof. Soheil Feizi
Image and Video Generative Models
Aug 2023 - Present
 - Enhancing the compositional abilities, localizing and editing knowledge, and improving image editing capabilities of Text-to-Image Generative Models, while exploring their interpretability and analyzing failure cases
- **Meta** Menlo Park, United State
Research Scientist Intern - Mentor: Wei Liu
Autoregressive Video Diffusion and World Models
Aug 2026 - Jan 2027
- **Adobe** San Jose, United State
Research Scientist Intern - Mentor: Samyadeep Basu
Controllable Image Editing (SliderEdit++ paper / project integrated into Adobe Firefly)
May 2026 - Aug 2026
- **Netflix** Los Gatos, United State
Research Scientist Intern - Mentor: Mahdi Kalayeh
Foundations of Diffusion and Flow Models (StructFlow paper)
Jan 2026 - May 2026
- **ByteDance - TikTok** San Jose, United State
Research Scientist Intern - Mentors: Jiacheng Pan, Matthew Gwilliam
Compositionality of Text-to-Image Models and Agentic LLMs (AgentComp paper)
Aug 2025 - Dec 2025
- **Pinterest** Remote, United State
Machine Learning Intern - Mentor: Eric Tzeng
Personalization of Text-to-Image Diffusion Models
May 2025 - Aug 2025
- **École Polytechnique Fédérale de Lausanne (EPFL)** Lausanne, Switzerland
Research Assistant - Summer Internship
Machine Learning and Signal Processing for Health Care (paper)
July 2022 - June 2023
- **Sharif University of Technology** Tehran, Iran
Research Assistant
Robustness in Deep Learning (paper1 and paper2)
Apr. 2021 - Jan. 2023
- **University of Arizona** Arizona, United States
Research Assistant - Summer Internship
3D Vision for Agriculture (paper)
July 2021 - Sep. 2021

PUBLICATIONS

- **SliderEdit: Continuous Image Editing with Fine-Grained Instruction Control**
A Zarei, S Basu, M Pournemat, S Nag, R Rossi, S Feizi
CVPR 2026 (Oral)
- **Localizing Knowledge in Diffusion Transformers**
A Zarei, S Basu, K Rezaei, Z Lin, S Nag, S Feizi
NeurIPS 2025
- **AgentComp: From Agentic Reasoning to Compositional Mastery in Text-to-Image Models**
A Zarei, J Pan, M Gwilliam, S Feizi, Z Yang
Under Review
- **Spatially-Grounded Flow Matching: Structured Source Distributions for Image Generation**
A Zarei, M Kalayeh
Under Review

- **SliderEdit++: Improved Slider Trajectories in Instruction-Based Image Editing**
A Zarei, S Basu, S Nag, F Derroncourt, B Wang, N Lipka, S Feizi *Under Review*
- **Understanding and Mitigating Compositional Issues in Text-to-Image Generative Models**
A Zarei, K Rezaei, S Basu, M Saberi, M Moayeri, P Kattakinda, S Feizi *Preprint*
- **Your Out-of-Distribution Detection Method is not Robust!**
M Azizmalayeri, A Soltani Moakar, A Zarei, R Zohrabi, MT Manzuri, MH Rohban *NeurIPS 2022*

Please refer to [Google Scholar](#) for other publications.

ACADEMIC SERVICES

- **Conference Reviewer**, CVPR/ECCV/NeurIPS/ICLR/ICML *2024 - Present*
- **Backend Lead**, [Sharif AI Challenge](#), Iran's Largest AI Competition *2021*
 - Managed the entire software development lifecycle, including architecture, solution design, and maintenance.

SELECTED ACADEMIC PROJECTS

- **Image Generative Models - PyTorch Implementation from Scratch:** [source-code](#)
A PyTorch implementation (from scratch) of various image generative models — Diffusion Models, GANs, VAEs, and Autoregressive Models — with detailed explanations, visualizations, and key mathematical formulas.
- **Selected Course Projects:**
Image Processing: [link](#) | Machine Learning: [link](#) | Modern Information Retrieval: [link](#) | Compiler Design: [link](#)
- **Others:** [Github](#)

WORK EXPERIENCE

- **Sotoon**
Software Engineer *Jan. 2022 - Apr. 2022*
 - A company providing Cloud and AI services
 - Project: Designed a Human Resource Management System
 - Technologies: Python, Django, Docker, HTML/CSS, Javascript, jQuery, Bootstrap
- **Snappfood**
Software Engineer *Aug. 2020 - Feb. 2021*
 - Leading online food ordering company in Iran
 - Back-End Developer in Menu Squad - Maintained menu APIs and designed new features
 - Technologies: PHP, Symfony, MySQL, Redis, Elasticsearch, Docker, RabbitMQ, HTML/CSS/Bootstrap, JS/jQuery

TEACHING ASSISTANT EXPERIENCE

- **Algorithms** *Fall 2023*
- **Machine Learning | Artificial Intelligence (2x) | Probability and Statistics** *Fall/Spring 2022*
- **Artificial Intelligence | Design of Algorithms (2x)** *Fall/Spring 2021*
- **Compiler Design | Data Structures and Algorithms | Advanced Programming** *Fall/Spring 2020*

RELEVANT COURSEWORK

- **Machine Learning:** 20/20
- **Image Processing:** 19.4/20
- ★ **Advanced Computer Graphics:** 4.0/4.0 (A⁺)
- ★ **Visual Learning and Recognition:** 4.0/4.0 (A⁺)
- ★ **Foundations of Deep Learning:** 4.0/4.0 (A⁺)
- ★ **Multimodal Foundation Models:** 4.0/4.0 (A⁺)
- **Stanford CS229 Machine Learning - CS231n DL for Computer Vision** (online, audited)
- **Generative Adversarial Networks (GANs) Specialization** (Coursera)

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C/C++, PHP, JavaScript, Golang
- **Machine Learning Libraries:** PyTorch, NumPy, Diffusers, Transformers, OpenCV, Scikit-Learn, Pandas, ...
- **Web Development & Database:** Laravel, Symfony, Django, NodeJS, ExpressJS, Wordpress, React, HTML/CSS, jQuery, Bootstrap, MySQL, Redis, MongoDB, Elasticsearch, RabbitMQ, ...
- **Miscellaneous:** Docker, Git, Linux

HONORS AND AWARDS

- Ranked **1st** in different Front & Back Development Contests (by SnappTrip, CodeCup, Edalat-Khaneh)
- Ranked **2nd** in the Provincial Chess Tournament
- Ranked among the top **1%** in the Nation-Wide University Entrance Exam

LANGUAGES

- **Persian:** native
- **English:** proficient (TOEFL 113: W30, S29, R27, L27)